B. MEXor Mixup

time limit per test 2 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1000 | Tags: bitmasks, greedy | https://codeforces.com/problemset/problem/1567/B

Alice gave Bob two integers a and b (a gt 0 and $b \geq 0$). Being a curious boy, Bob wrote down an array of **non-negative** integers with MEX value of all elements equal to a and XOR value of all elements equal to b.

What is the *shortest* possible length of the array Bob wrote?

Recall that the MEX (Minimum EXcluded) of an array is the minimum non-negative integer that does **not** belong to the array and the XOR of an array is the bitwise XOR of all the elements of the array.

Input

The input consists of multiple test cases. The first line contains an integer t ($1 \leq t \leq 5 \cdot 10^4$) — the number of test cases. The description of the test cases follows.

The only line of each test case contains two integers a and b ($1 \leq a \leq 3 \cdot 10^5$; $0 \leq b \leq 3 \cdot 10^5$) — the MEX and XOR of the array, respectively.

Output

For each test case, output one (positive) integer — the length of the shortest array with MEX a and XOR b. We can show that such an array always exists.

Example

Input
5 1 1 2 1 2 0 1 10000 2 10000

Output
3 2 3 2 3

Note

In the first test case, one of the shortest arrays with MEX 1 and XOR 1 is [0, 2020, 2021].

In the second test case, one of the shortest arrays with MEX 2 and XOR 1 is [0, 1].

It can be shown that these arrays are the shortest arrays possible.